

CSA6102 – DIGITAL FORENSICS

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EXP 1

1. Write a Python program to demonstrate Locard's Exchange Principle using file timestamps (metadata trace demo).

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# =====
# Locard's Exchange Principle - Metadata Trace Demo
# Google Colab Compatible
# =====

import os
import time
from datetime import datetime

filename = "digital_evidence.txt"

# -----
# Step 1: Create the evidence file
# -----
with open(filename, "w") as file:
    file.write("Original Digital Evidence\n")

print("Evidence file created.")

# Wait so timestamps will differ
time.sleep(3)

# -----
# Step 2: Capture original metadata
# -----
original = os.stat(filename)

print("\n===== ORIGINAL METADATA =====")
print("File Name :", filename)
print("Last Access Time :", datetime.fromtimestamp(original.st_atime))
print("Last Modified Time :", datetime.fromtimestamp(original.st_mtime))
print("Metadata Size :", original.st_size, "bytes")

# -----
# Step 3: Simulate tampering
# -----
print("\nSimulating file modification...")

with open(filename, "a") as file:
    file.write("Unauthorized modification detected.\n")

# Wait a little
time.sleep(2)

```



```
# -----
# Step 4: Capture new metadata
# -----
updated = os.stat(filename)

print("\n===== UPDATED METADATA =====")
print("File Name :", filename)
print("Last Access Time :", datetime.fromtimestamp(updated.st_atime))
print("Last Modified Time :", datetime.fromtimestamp(updated.st_mtime))
print("Metadata Size :", updated.st_size, "bytes")

# -----
# Step 5: Compare metadata
# -----
print("\n===== METADATA COMPARISON =====")

if original.st_mtime != updated.st_mtime:
    print("Modification Time Changed")
    print("Trace Detected: File has been modified.")
else:
    print("No modification detected.")

if original.st_size != updated.st_size:
    print("File Size Changed")
else:
    print("File Size Unchanged")

# -----
# Step 6: Display file contents
# -----
print("\n===== FILE CONTENT =====")

with open(filename, "r") as file:
    print(file.read())

# -----
# Step 7: Demonstrate Locard's Principle
# -----
print("\n===== CONCLUSION =====")
print("Locard's Exchange Principle states:")
print("'Every contact leaves a trace.'")

print("\nIn this demonstration:")
print("- The file was accessed and modified.")
print("- The metadata (timestamps and size) changed.")
```

OUTPUT :

*** Evidence file created.

===== ORIGINAL METADATA =====

File Name : digital_evidence.txt
Last Access Time : 2026-07-11 04:09:35.920771
Last Modified Time : 2026-07-11 04:09:35.921771
Metadata Size : 26 bytes

Simulating file modification...

===== UPDATED METADATA =====

File Name : digital_evidence.txt
Last Access Time : 2026-07-11 04:09:35.920771
Last Modified Time : 2026-07-11 04:09:38.925992
Metadata Size : 62 bytes

===== METADATA COMPARISON =====

Modification Time Changed
Trace Detected: File has been modified.
File Size Changed

===== FILE CONTENT =====

Original Digital Evidence
Unauthorized modification detected.

===== CONCLUSION =====

Locard's Exchange Principle states:
"Every contact leaves a trace."

In this demonstration:

- The file was accessed and modified.
 - The metadata (timestamps and size) changed.
 - These changes serve as digital traces that can be examined during forensic investigations.
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